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TRIBE Stimulus Discovery as a Bounded Autoresearch Case Study

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Date: 2026-04-28

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Abstract

When a person hears a spoken narrative versus a spoken arithmetic problem, what dimension of the auditory stimulus separates them — and is that same dimension encoded in human language cortex? More broadly, across many candidate cognitive contrasts (story vs arithmetic, social agency vs mechanical motion, theory-of-mind vs physical reasoning, speech vs non-speech sound), which stimulus axes deserve resource-intensive subject-level fMRI validation in the first place, and which are pipeline artifacts that should not be tested at all? These are the questions we ask of a state-of-the-art multimodal brain-encoding model.

We address them by using TRIBE v2 (d'Ascoli et al., 2025) as a computational screening tier: an offline brain-encoding substrate that predicts `fsaverage5` cortical responses to video, audio, and text before subject-level validation is scheduled. BR converts each candidate contrast into model inputs, extracts model representations and predicted cortical responses, tests whether the conditions separate beyond permutation baselines, and then applies fixed rules to decide whether the contrast should advance, be revised, or stop. Across seven contrasts, the campaign yields one positive axis, three candidate axes requiring further controls, and three failures attributable to how stimuli were represented for the model. The strongest result is converted into a registered scientific hypothesis (H1) and a methodological hypothesis (H2).

H1 (registered). The story-audio versus math-audio separation observed in late TRIBE encoder attention modules (`encoder.layers.{10,12,14}.1`) reflects a high-level narrative/semantic processing dimension that is also encoded in left-lateralized human language cortex — distinct from low-level acoustic structure, arithmetic-specific computation, and generic task-difficulty. The supporting evidence is item-level and model-internal: original 5-vs-5 separation (score 15.15, exact $p = 0.0079$); expanded 20-vs-20 validation (score 10.56, plus-one MC $p \sim 5e-5$, $N = 20,000$); held-out 41-item validation (plus-one $p \sim 2e-4$, Bonferroni $p \sim 4e-4$); source-family sensitivity (Bonferroni $p = 0.0434$); acoustic-covariate residualization (plus-one $p \sim 5e-5$); and a predeclared late-vs-early encoder layer-family confirmation ($T_{\text{late_minus_early}} = 0.859$, plus-one $p \sim 5e-5$). H1 makes four falsifiable predictions for the next decisive subject-level experiment in observed HCP LANGUAGE fMRI (P1–P4, §4).

H2 (methodological). A bounded autonomous loop, when constrained by checks that follow-ups change the stimulus inputs, by trajectory-level hypothesis definitions, and by branch-specific stopping rules, produces hypothesis classes whose distribution is informative about both biology and pipeline limits. In particular, the rate of stimulus-representation failures is a measurable property of the materialization layer, not of the underlying neuroscience.

Boundary of claim. The current HCP language evidence is item-level and model-internal. No subject/run-aligned HCP LANGUAGE BOLD, CIFTI, NiftI, beta, COPE, ZSTAT, or stat-map targets are present in the workspace, and the predicted-fMRI fold-stability bridge between the expanded20

and heldout21 stimulus folds is not stable ($\bar{r} = 0.098$, permutation $\bar{p} = 0.442$). The predictions of H1 are stated for the next experiment, not as already-observed results.

Abbreviations

- **A1** — primary auditory cortex
- **AG** — angular gyrus
- **ATL** — anterior temporal lobe
- **BOLD** — Blood-Oxygen-Level-Dependent fMRI signal
- **CIFTI** — Connectivity Informatics Technology Initiative file format
- **COPE / ZSTAT** — Contrast of Parameter Estimates / standardized z-statistic map (FSL)
- **fsaverage5** — FreeSurfer average cortical surface, ~20,484 vertices
- **HCP** — Human Connectome Project
- **HCP-MMP** — HCP Multi-Modal Parcellation
- **IBC** — Individual Brain Charting
- **IFG** — inferior frontal gyrus
- **MFCC** — Mel-Frequency Cepstral Coefficients
- **OLS** — Ordinary Least Squares
- **ROI** — Region of Interest
- **RSVP** — Rapid Serial Visual Presentation
- **STG** — superior temporal gyrus
- **ToM** — Theory of Mind
- **TRIBE** — TRImodal Brain Encoder (d'Ascoli et al., 2025)
- **V1** — primary visual cortex

1. Introduction

Brain-encoding models are usually evaluated by asking whether model features or predicted responses explain known neural measurements. The scientific motivation here is broader. We ask whether a brain-encoding model can seed a self-driven discovery loop that proposes stimulus contrasts, tests whether the conditions separate in model or predicted-response space, and converts each candidate contrast into either a bounded hypothesis or a concrete redesign instruction.

The core object of the study is the relationship among stimuli, model representations or layers, and predicted or observed neural responses. The analysis follows a simple chain: a stimulus item is recorded in a manifest, converted into an input event for TRIBE, represented as model features or predicted cortical responses, scored for condition separation, and then assigned a next step. This schema matters because the loop does not treat a single high score as a hypothesis. In this report, a *branch* is the full test record for one candidate contrast: the input stimuli, the score pattern, any follow-up or validation result, and the decision to advance, revise, or stop. On top of that, the campaign as a whole is responsible for delivering a *scientific hypothesis*, a falsifiable claim about the brain, which is stated explicitly in §4.

This framing also makes negative outcomes interpretable. A weak result can mean several different things: the biological axis may be absent from the current stimulus set, task-defining information may have been lost when stimuli were converted into model inputs, the representation may be mismatched to the domain, or the follow-up may have repeated the same input set. The goal of the report is therefore not to claim that every score is a neural effect. The goal is to show how a bounded autonomous loop sorted candidate contrasts into positive axes, findings that need further controls, failures caused by stimulus representation in the current pipeline, and operational rules for better future experiments. The strongest branch is then converted into a registered scientific hypothesis.

Figure 1 gives the conceptual setup. It should be read as a schematic of the workflow, not as evidence for any particular neural effect.

1.1 Related work

Two literatures bear directly on this report.

Brain-encoding models with audio-language stimuli include linearized encoding of natural-speech stimuli (Huth et al., 2016), language-model-based prediction of language-evoked responses (Schrimpf et al., 2021; Caucheteux & King, 2022), and the multimodal extensions that motivate TRIBE itself (d’Ascoli et al., 2025). Within HCP language specifically, story-vs-math contrasts have been used to localize left-lateralized language regions following the Barch et al. (2013) protocol; story blocks engage classical perisylvian language cortex (STG, IFG, ATL, AG), while math blocks

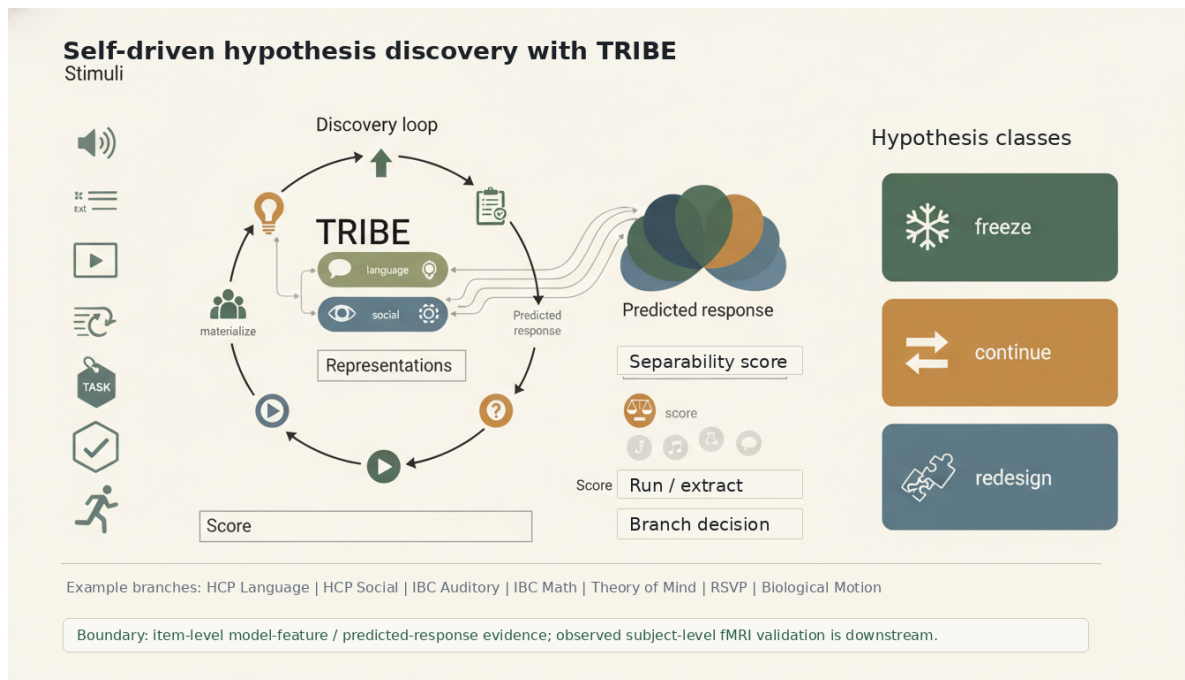


Figure 1: Conceptual schematic

engage parietal/IPS cortex more strongly. H1 below is consistent with that literature but is stated as a model-anchored prediction rather than a redescription of it.

Autonomous scientific discovery with LLM agents has progressed rapidly in the past two years: AI Scientist (Lu et al., 2024), FunSearch (Romera-Paredes et al., 2024), and related autoresearch systems propose, run, and write up experiments end-to-end. Most published systems target ML benchmarks or mathematical conjectures. The contribution of the current campaign is methodological: applying the same loop discipline to a stimulus-design and brain-encoding setting where the unit of progress is *which stimulus contrast deserves resource-intensive subject-level fMRI validation*, and where the dominant failure modes are stimulus-materialization issues rather than reasoning errors.

2. Methods

2.1 Model and prediction artifacts

The model engine is TRIBE v2, the TRImodal Brain Encoder described by d’Ascoli et al. (2025) and distributed as the Hugging Face model `facebook/tribev2`. The model card describes TRIBE v2 as a multimodal brain-encoding model that predicts fMRI responses to video, audio, and text on the `fsaverage5` cortical mesh. In this campaign it was loaded with `checkpoint_name=best.ckpt` through `tribev2.TribeModel.from_pretrained(...)`. The runtime prediction path was inspected through the TRIBE sweep script and the Brain Researcher tool wrapper. Stimuli were represented as manifest rows and runtime events. Depending on the branch, events could include text, audio, video, and task-condition metadata. The inspected checkpoint declares modality feature dimensions `text=(2,3072)`, `audio=(2,1024)`, and `video=(2,1408)`, with `n_outputs=20484` and `n_output_timesteps=100`.

The primary prediction artifacts used by the loop are `embedding_rows.jsonl`, `embeddings_matrix.npy`, per-item matrix files, and per-item embedding files. Rows record fields such as `item_id`, `condition`, `task_id`, `n_vertices`, `segment_count`, and runtime metadata. The surface outputs used in the current figures are `fsaverage5` predicted-response vectors with 20,484 vertices. These are model-predicted surface responses, not observed subject-level BOLD maps.

Layer-feature sidecars were added in a later pass by rerunning the checkpoint with hooks. The confirmed hook families include `projectors.audio`, `projectors.text`, `selected_encoder.layers.{0,2,4,10,12,14}.1` modules, and related encoder outputs. These sidecars support item-level layer-family claims for HCP language, but they do not provide observed fMRI evidence.

2.2 Stimulus materials

HCP language. The language task (Barch et al., 2013) presents approximately 30-second auditory blocks of two types. *Story* blocks present briefly adapted Aesop fables narrated by a single speaker, followed by a two-alternative comprehension question. *Math* blocks present spoken arithmetic problems (addition and subtraction of multi-digit numbers) followed by a two-alternative answer choice. Both block types are auditory- only and matched on overall block duration. In this campaign the original loop selected 5 story and 5 math items (with 339 vs 186 underlying segments in round 1); the expanded validation used 20 of each; the held-out validation used 41 additional items not seen during the original branch selection.

HCP social. Social blocks present Heider-and-Simmel-style geometric- shape animations depicting agentic interaction; mechanical-motion blocks present similar shapes moving according to physical/random trajectories without apparent agency. The current campaign packaged 10 social and 7 mechanical items in round 1, then balanced to 5/5 in round 2.

IBC auditory, math, ToM. Stimuli are drawn from the Individual Brain Charting battery as packaged in the local manifest set. Auditory contrasts compare speech against animal/nature/tools/voice/music sound categories; math contrasts compare arithmetic-principle blocks against lexical control conditions; ToM contrasts compare belief-question against physical-question probes following established false-belief paradigms.

RSVP language and biological motion. The current materialization collapsed task-defining temporal structure: RSVP word-by-word timing was reduced to static condition labels, and biological-motion stimuli were averaged across frames rather than represented as motion energy. These are flagged as packaging-sensitive failures rather than null biological results.

2.3 Autonomous branch loop

Each branch starts from a scientific contrast — for example HCP language `story_audio` vs `math_audio`, HCP social `social_animation` vs `mechanical_motion`, IBC auditory `speech` vs `sound` controls, IBC math `arithmetic-principle` vs `control` conditions, ToM `belief` vs `physical` questions, RSVP language, or biological motion. The loop materializes a stimulus manifest, runs TRIBE, scores condition separation, proposes a follow-up or terminal decision, and routes the branch to freeze, continue, kill, or redesign.

An important methodological guardrail is **manifest-delta validation**. A follow-up is considered scientifically meaningful only if the condition signature actually changed. This rule was necessary because earlier branches could appear to make progress while repeating the same materialized contrast or dropping the wrong control condition.

2.4 Score definition

For a contrast with positive condition set P and negative condition set N , let $\mathbf{c}_P$ and $\mathbf{c}_N$ be the corresponding centroid vectors in the chosen feature space (raw predicted response, residualized response, or layer-feature embedding). Define:

- $\text{diff_norm} = ||\mathbf{c}_P - \mathbf{c}_N||_2$
- $\text{cosine_gap} = 1 - \cos(\mathbf{c}_P, \mathbf{c}_N)$, where $\cos$ is cosine similarity

The branch score is then:

$$\text{score} = \text{diff_norm} * \max(\text{cosine_gap}, 1e-6)$$

The $1e-6$ floor prevents pathological behavior when centroids are nearly collinear. The score is a prioritization statistic, not a p-value or neural effect size; significance is established separately via the permutation procedures of §2.6. Scores across raw predicted responses, residualized features, and layer-feature spaces are not numerically interchangeable.

2.5 Layer-family definition

The TRIBE v2 checkpoint exposes 24 transformer encoder layers indexed 0–23 plus modality projectors. For the layer-family confirmatory test (§2.6), three families are defined a priori from the architecture:

- **Late encoder attention** — attention sublayers `encoder.layers.{10,12,14}.1`
- **Early encoder attention** — attention sublayers `encoder.layers.{0,2,4}.1`
- **Projectors** — `projectors.audio, projectors.text`

The predeclared statistic is

`T_late_minus_early = mean(scores_late) - mean(scores_early)`

with positive values supporting a late-locus hypothesis. Per-layer scores are reported in Table 3 (§3.3).

2.6 Statistical validation

Permutation procedure. All Monte Carlo permutation tests use $N = 20,000$ random label permutations with the standard plus-one correction $p = (1 + \#\{T_{\text{perm}} \geq T_{\text{obs}}\}) / (1 + N)$. Random seeds are branch-specific and recorded in each `*_validation.json` file alongside the test. The exact item-label test for the original 5-vs-5 contrast enumerates all $(10 \text{ choose } 5) = 252$ partitions.

Source-family sensitivity. HCP language stimuli are drawn from multiple narrator/source families (Aesop fables narrated by different speakers; arithmetic problems generated from different templates). The source-family sensitivity test repeats the contrast permutation test stratified by source family and applies Bonferroni correction across the family count. This guards against the contrast being driven by a single narrator or template.

Acoustic covariate adjustment. Predicted responses and layer features are residualized against per-item duration, RMS loudness, and segment count via OLS regression before permutation. This is a *limited* acoustic feature set (see §7); F0, spectral centroid, speech rate, and MFCC summaries are not available in the current covariate sidecar and are scheduled for the acoustic-robustness prediction (P3, §4.2).

Layer-family confirmatory test. The predeclared statistic is `T_late_minus_early` (§2.5). The supporting locked manifest builder and validator scripts were committed together at git commit `7f1c89b1` ("Run HCP language layer-family confirmatory validation"), which serves as the predeclaration anchor for this report; locked specifications are recorded in `docs/operations/locked_followup_manifest_specs_2026-04-26.md`. We note that strict pre-registration would require an earlier commit isolating the predicted statistic from its evaluation; this report treats `7f1c89b1` as a soft pre-registration anchor and the H1 predictions in §4 as the strict pre-registration for the next decisive experiment.

2.7 Neural validation boundary

The workspace does not currently contain subject/run-aligned observed HCP LANGUAGE fMRI targets. Specifically, no aligned BOLD, CIFTI, NIfTI, beta, COPE, ZSTAT, or stat-map target package is present in the current local or VM artifacts, and the prediction rows do not expose `subject_id` or `fold_id`. Subject-level neural validation is therefore a required future step and is the decisive experiment defined in §5. The surface maps shown in this report are predicted-response diagnostics, not observed activation maps.

2.8 Compute and resources

TRIBE v2 inference for all reported branches was run on the closed-loop VM under `/home/ubuntu/tribe_encoding/` against the `facebook/tribev2` checkpoint (`best.ckpt`). Per-branch inference is dominated by checkpoint forward passes over the materialized manifest items; total inference compute across all reported branches is on the order of single-digit GPU-hours on a single CUDA device, with exact wall-time logs in the per-branch `predictions/` directories. Validation and permutation scripts run on CPU and complete in seconds-to-minutes per branch. Loop-controller LLM token usage is recorded in the unified ledger but is not in the critical path of any reported scientific claim.

3. Results

3.1 A computational screening tier sorted branches into distinct hypothesis classes

Subject-level encoding against observed BOLD data is resource-intensive, so not every candidate contrast should advance directly to that tier. In this campaign, BR used TRIBE v2 as a computational screening tier: a model-based cortical substrate used to evaluate candidate contrasts before subject-level validation was scheduled. For each proposed contrast, BR converted stimuli into model inputs, ran TRIBE to obtain model representations or predicted `fsaverage5` responses, scored condition separation, and applied predeclared branch decisions. This screening tier was consequential because it changed which claims advanced.

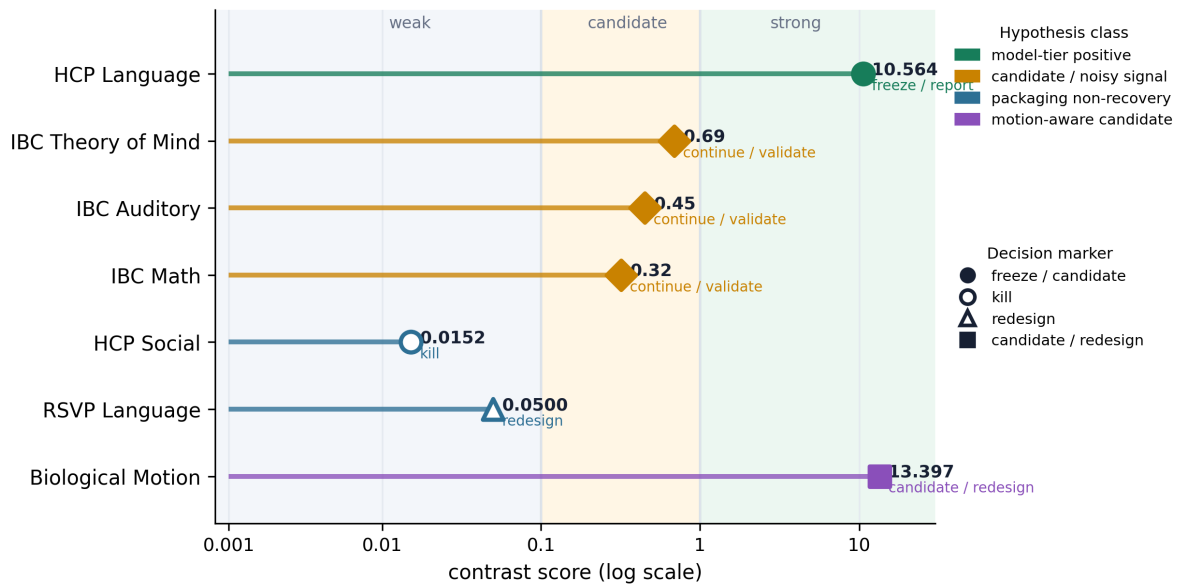
The branch landscape shows that the loop did not produce one homogeneous result. HCP language became a robust positive axis and was promoted to a registered subject-level hypothesis. Math, auditory, and ToM retained nonzero but incomplete signals after stimulus-input fixes and therefore require additional controls. HCP social, RSVP language, and biological motion became current-pipeline failures, meaning that the present way of representing those stimuli for TRIBE was not sufficient to support a biological claim. Figure 2 summarizes the terminal or current state of each branch using the latest audited score and decision; Table 2 gives the underlying numbers.

Table 2. Branch outcome master table. Score is the §2.4 statistic. `n_pos / n_neg` are item counts in the materialized manifest; `-` indicates counts not recorded in the figure data table. Class taxonomy: *real_positive* (advanced to a reportable claim); *candidate_noisy* (requires follow-up before claim); *packaging_failure* (current-pipeline non-recovery caused by stimulus representation, requiring redesign).

Branch	Round	Contrast	Score	<code>n_pos / n_neg</code>	Decision	Class
HCP language	1	story_audio vs math_audio	15.154	339 / 186	freeze	real_positive
HCP language	2	story_audio vs math_audio (expanded20)	10.564	20 / 20	freeze	real_positive
HCP social	1	social_animation vs mechanical_motion	0.0349	10 / 7	follow-up	candidate_noisy
HCP social	2	social_animation vs mechanical_motion (balanced)	0.0152	5 / 5	kill	packaging_failure
IBC math	1	arithprin vs control_lexical	0.32	12 / 12	tighten	candidate_noisy
IBC ToM	1	belief_question vs physical_question	0.69	10 / 10	follow-up	candidate_noisy
IBC auditory	1	speech vs other_sounds	0.45	8 / 24	follow-up	candidate_noisy
IBC RSVP	1	rsvp_target vs rsvp_distractor	0.05	—	redesign	packaging_failure
IBC biological motion	1	intact vs scrambled	0.0019	—	redesign	packaging_failure

Figure 2. Self-driven experiments sort stimulus contrasts into hypothesis classes

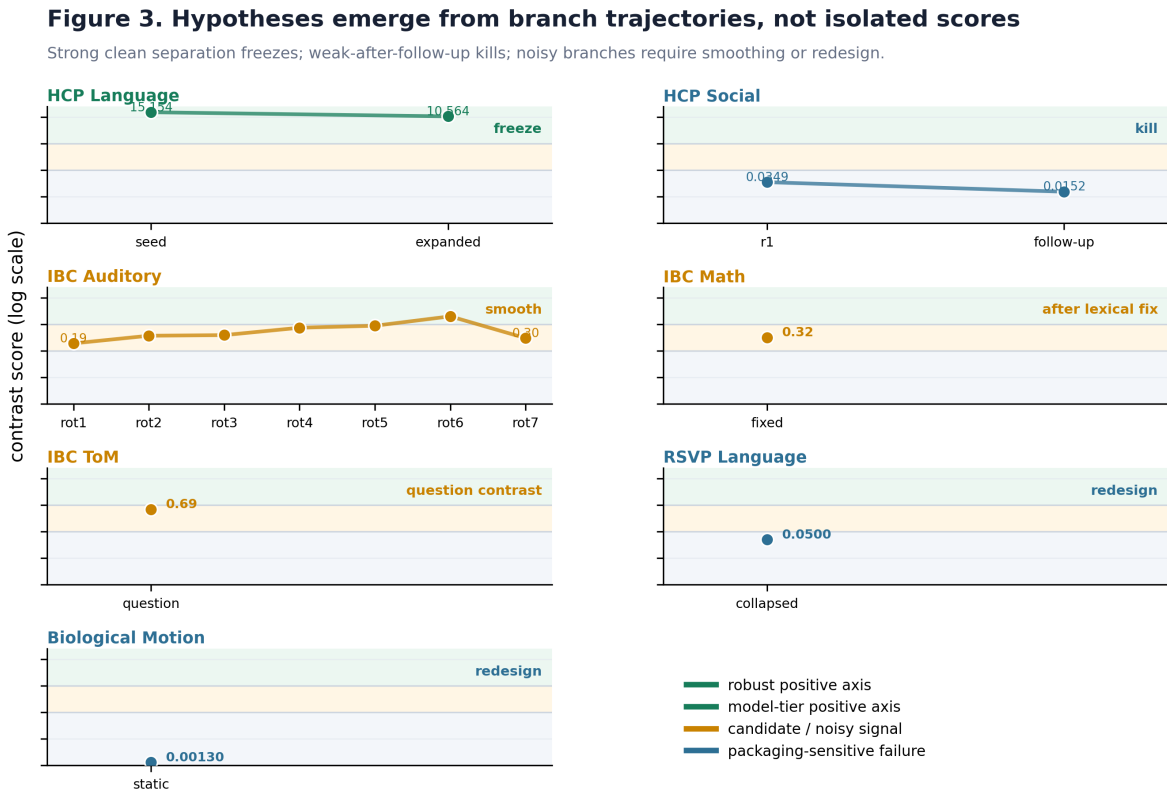
Final branch states show 1 model-tier positive, 3 standard candidates, 2 packaging failures, and 1 motion-aware redesign candidate.



Score is an automated representation / predicted-response separability measure, not a p-value or direct neural effect size.

Figure 2: Branch outcome landscape

Branch decisions depend on the full test record, not isolated scores. HCP language showed a large separation and advanced. HCP social was weak initially and weaker after a focused follow-up, supporting a stop-and-redesign decision in the current pipeline. Auditory showed nonzero but unstable item-rotation behavior, which argues for smoothing and larger item coverage rather than immediate reporting. Figure 3 makes this trajectory logic explicit.



Single-point branches mark terminal/current branch states; no-op/fix steps without audited scores are not assigned fake values.

Figure 3: Branch trajectories

3.2 HCP language was the strongest positive item-level axis

HCP language `story_audio` vs `math_audio` was the strongest positive branch. The original selected 5-vs-5 contrast had score 15.1541 with exact item-label permutation $p = 0.00794$. The expanded 20-vs-20 validation remained strong, with score 10.5640 and plus-one Monte Carlo $p = 4.99975e-05$. The held-out 41-row item test remained significant with score 0.4108, plus-one $p = 0.00019999$, and Bonferroni-adjusted $p = 0.00039998$. The effect was weaker but still significant when the test accounted for stimulus source families, with corrected $p = 0.0434$. It also remained significant after adjusting for the available acoustic covariates (duration, loudness, and segment count), with plus-one $p = 4.99975e-05$ and Bonferroni $p = 0.00014999$. Table 1 collects the full validation ladder.

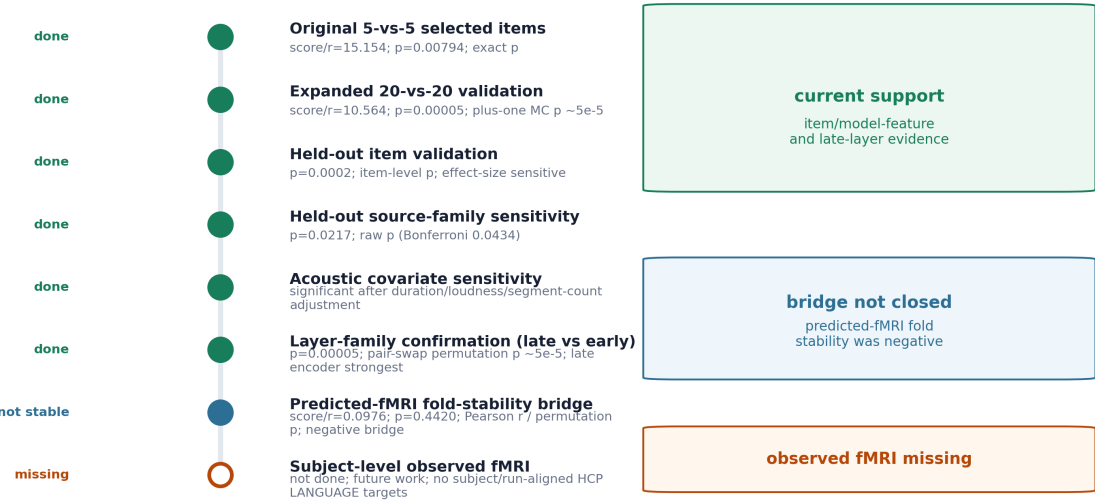
Table 1. HCP language validation ladder. Tiers 1–6 support the item-level and model-feature claim; tier 7 is a negative result that prevents upgrading to a stable predicted-neural-map claim; tier 8 is the missing decisive evidence.

Tier	Test	Score / stat	p-value	p-label	Status
1	Original 5-vs-5 selected items	15.154	0.00794	exact (252 partitions)	done – strong
2	Expanded 20-vs-20 validation	10.564	$\sim 5e-5$	plus-one MC, $N = 20,000$	done – strong
3	Held-out 41-item validation	0.411	$\sim 2e-4$	plus-one MC, $N = 20,000$	done – significant
4	Held-out source-family sensitivity	—	0.0217 raw / 0.0434 Bonf	Bonferroni-corrected	done – significant
5	Acoustic covariate (dur / loud / segcount)	—	$\sim 5e-5$ / $1.5e-4$	plus-one MC / Bonf	done – strong
6	Layer-family confirmation (late vs early)	$T = 0.859$	$\sim 5e-5$	plus-one MC pair-swap	done – strong
7	Predicted-fMRI fold-stability bridge	$r = 0.098$	0.442	Pearson / permutation	done – not stable
8	Subject-level observed fMRI	—	—	—	missing – required (\$5)

The evidence ladder is also rendered as Figure 8.

Figure 8. Neural claim boundary for HCP language

The strongest current claim is item-level model-feature / predicted-response evidence; observed subject-level fMRI remains open.



Do not interpret this ladder as an observed activation map. It shows which neural-evidence tiers are complete, unstable, or missing.

Figure 4: Neural validation ladder

3.3 Late TRIBE encoder representations carried the HCP language separation

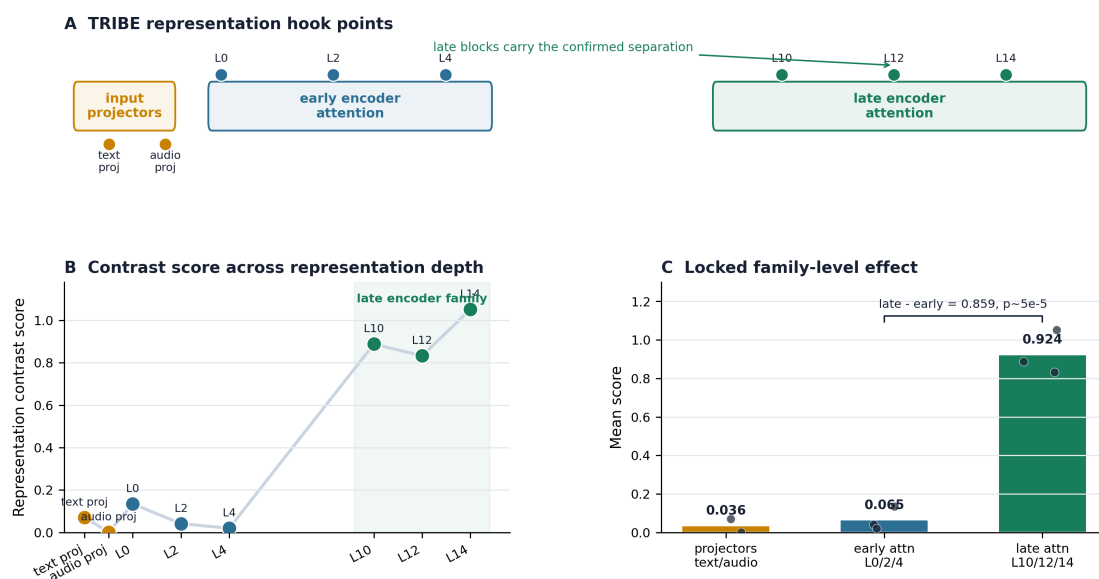
The layer-feature analysis refined where the HCP language separation appeared inside TRIBE. The separation was not strongest in raw modality-projector features. It was strongest in late TRIBE encoder attention representations. The locked layer-family test confirmed the predeclared direction: late encoder attention modules exceeded early encoder modules, with $T_{\text{late_minus_early}} = 0.8590$ and plus-one permutation $p = 4.99975e-05$. The full per-layer profile is Table 3 and Figure 11.

Table 3. Layer-family scores on the HCP language story-vs-math contrast. $\text{mean}(\text{late}) = 0.924$, $\text{mean}(\text{early}) = 0.066$, $T_{\text{late_minus_early}} = 0.859$, plus-one MC pair-swap $p \sim 5e-5$ ($N = 20,000$).

Family	Layer	Score
Late encoder attention	encoder.layers.14.1	1.0517
Late encoder attention	encoder.layers.10.1	0.8884
Late encoder attention	encoder.layers.12.1	0.8329
Early encoder attention	encoder.layers.0.1	0.1347
Early encoder attention	encoder.layers.2.1	0.0414
Early encoder attention	encoder.layers.4.1	0.0199
Projectors	projectors.text	0.0707
Projectors	projectors.audio	0.0018

Figure 11. HCP language representation profile in TRIBE

Story-audio vs math-audio separation is strongest in late encoder attention features, not projector-level features.



Representation-space figure only. It summarizes locked layer scores; raw item x feature matrices are not local, so item-level PCA/UMAP is deferred. This is not observed fMRI activation.

Figure 5: HCP language TRIBE representation profile

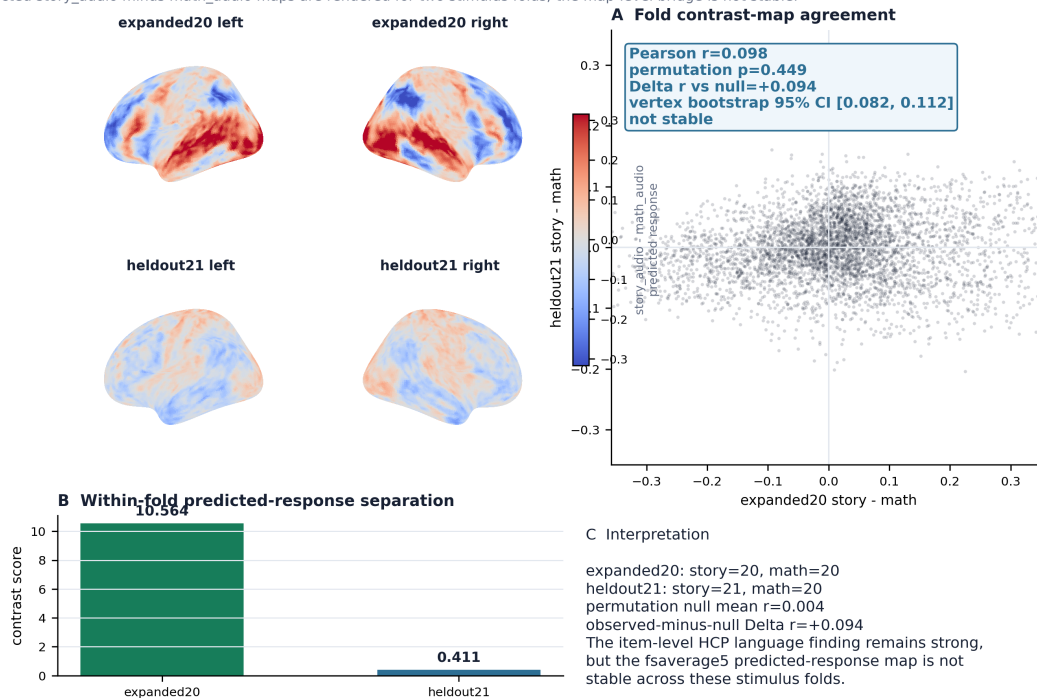
This result supports a bounded claim: the story-vs-math audio axis is more separable in late multimodal TRIBE encoder representations than in early or projector-level features. It does not by itself establish observed neural activation, ROI specificity, or subject-level consistency; those are addressed by the predictions in §4.

3.4 The HCP language predicted-response map bridge did not validate a stable neural map

The available HCP language prediction matrices can be rendered as `fsaverage5` predicted-response maps. However, the map-level bridge between `expanded20` and `heldout21` stimulus folds was not stable. The story-minus-math contrast maps had Pearson $r \approx 0.098$ and permutation $p \approx 0.442$. Figure 13 shows both the surface maps and the fold-level scatter. This is a limitation figure: it prevents upgrading the item-level finding to a stable predicted neural-response map claim, and it makes P2 (§4.2) the load-bearing test of H1.

Figure 13. HCP language predicted `fsaverage5` response bridge

TRIBE-predicted `story_audio` minus `math_audio` maps are rendered for two stimulus folds; the map-level bridge is not stable.



Predicted response only. This is not observed subject fMRI; the negative bridge prevents upgrading the claim to a stable neural-response map.

Figure 6: HCP language predicted `fsaverage5` bridge

3.5 HCP social was not recovered by the current pipeline

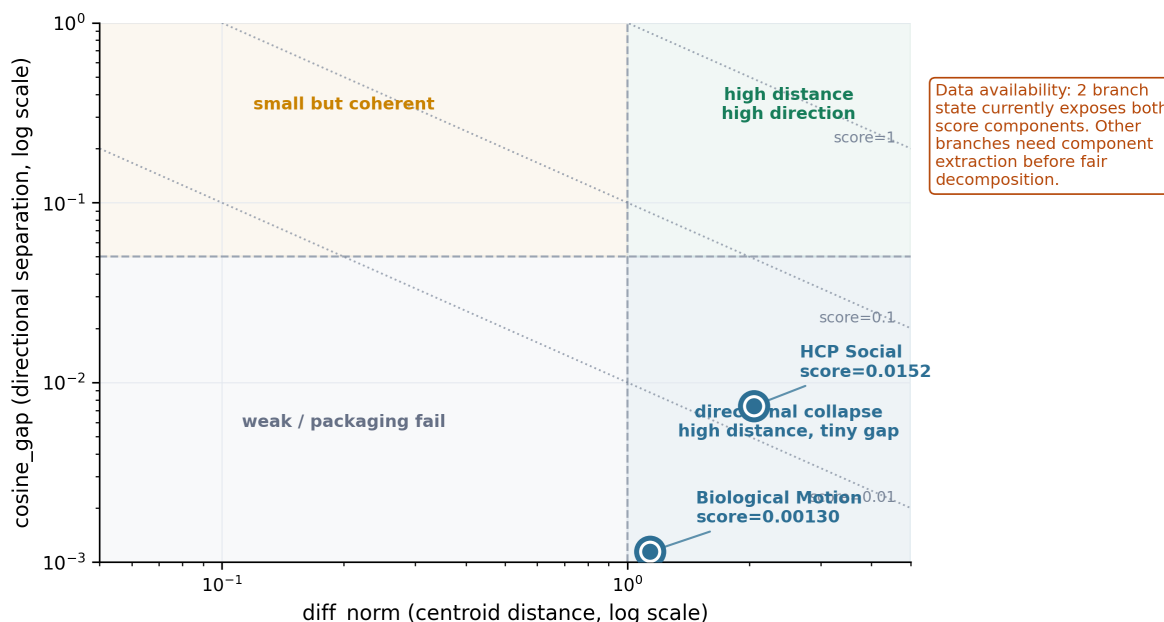
The HCP social branch did not support a positive claim. The first social branch score was 0.03494 with exact $p = 0.07937$. A focused follow-up dropped to score 0.01523 with exact $p = 0.24603$. All-item HCP social validation remained non-significant, with score 0.01148, exact $p = 0.10068$, and Bonferroni $p = 0.20136$. The correct interpretation is that the current packaging and

representation setup did not recover social-animation vs mechanical-motion separation. It is not evidence that social cognition lacks neural signal.

The score-decomposition plot in Figure 4 clarifies why this matters. HCP social did not fail only because the centroid distance was zero. It failed because directional separation was tiny after a valid follow-up. This distinction turns a weak score into a more informative redesign instruction.

Figure 4. Low score can mean directional collapse, not zero distance

Score = $\text{diff_norm} \times \max(\text{cosine_gap}, 1e-6)$. HCP Social has distance but almost no stable direction.



Component-level decomposition is available for HCP Social follow-up only; missing components are not imputed.

Figure 7: Score decomposition

3.6 Manifest changes were necessary for valid branch interpretation

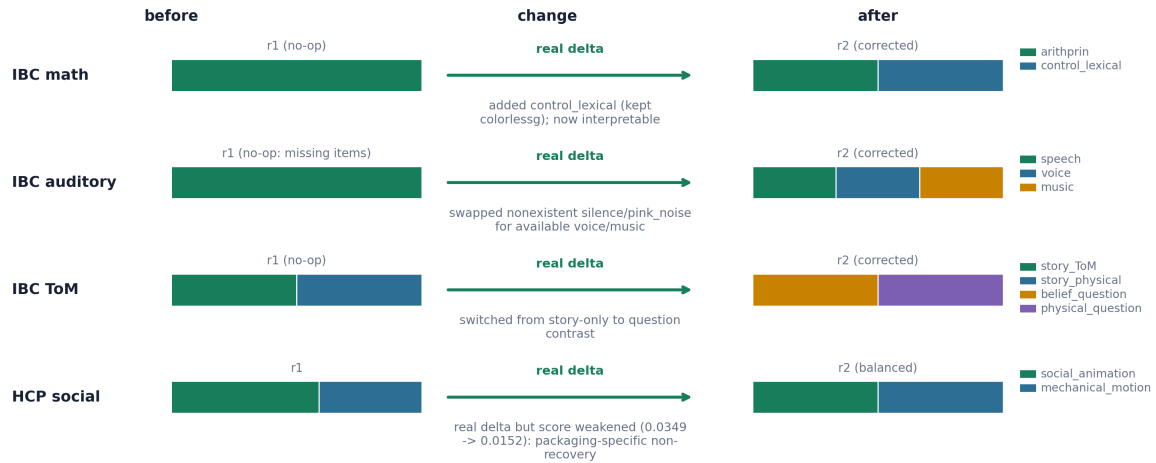
Several branches became interpretable only after checking whether the follow-up actually changed the manifest. Math required retaining the lexical control condition rather than dropping it. Auditory required using available controls such as voice and music rather than requesting nonexistent silence/pink-noise controls. ToM required moving from a story-only plateau to belief-vs-physical question contrasts. HCP social had a real balanced follow-up and still weakened, which supports stopping that branch in the current setup. Figure 5 summarizes these condition-signature changes.

3.7 Candidate and packaging-sensitive branches became next-experiment hypotheses

Math, auditory, and ToM are not final discoveries, but they became candidate hypotheses after materialization fixes. Math showed a nonzero candidate signal after retaining lexical controls.

Figure 5. A follow-up only counts if the condition signature changes

Before/after condition bars show when controller actions became real stimulus interventions.



Counts are condition-signature summaries for interpretability; they are not new validation sample-size claims.

Figure 8: Condition signature changes

Auditory showed nonzero but noisy behavior, implying larger item budgets and acoustic controls. ToM improved when the contrast moved to belief-vs-physical questions rather than story-only materialization. RSVP language and biological motion remained packaging-sensitive failures because the current materialization did not preserve word timing, probe structure, or dynamic motion energy.

Figure 6 converts these branch outcomes into findings, hypotheses, and next experiments. Figure 7 turns the same logic into an experimental redesign roadmap.

Figure 6. The loop output is a hypothesis map, not just a score table

Each branch is mapped from finding to bounded hypothesis, terminal decision, and next experiment.

Branch	Finding	Hypothesis	Decision	Next experiment
HCP Language	Strong story-vs-math audio separation	Robust auditory language/story axis in current TRIBE evidence	freeze / report	Subject/run-aligned observed HCP LANGUAGE validation; cross-modal text/audio check
HCP Social	Weak after valid follow-up	Current setup misses social-motion axis	kill / redesign	Motion-energy matched social/mechanical videos with motion-aware features
IBC Auditory	Nonzero but noisy	Speech/natural-sound axis is sample-sensitive	continue / validate	Larger item budget plus acoustic voice/music/nonspeech controls
IBC Math	Nonzero after lexical fix	Arithmetic principle separable only with lexical controls kept	continue / validate	Difficulty, lexical, visual, and syntactic-control validation
IBC ToM	Question contrast improved interpretability	Belief-vs-physical question axis is more promising than story-only	continue / validate	Matched belief vs physical question battery
RSVP Language	Low under flattened packaging	Timing/probe structure likely lost	redesign	Timing/probe-preserving RSVP trial manifest
Biological Motion	Near-zero under current representation	Motion structure not captured by static/motion-poor packaging	redesign	Dynamic intact vs scrambled motion videos with motion-aware representation

Figure 9: Finding-hypothesis-next-experiment matrix



Figure 10: Stimulus redesign roadmap

3.8 Neural and ROI figures should be read as target maps or predicted-response diagnostics

The current report includes neural-style figures because the model outputs live in `fsaverage5` predicted-response space. These figures are useful for planning and diagnostics, but they are not observed fMRI activation maps. Figure 9 shows ROI systems that should be targeted in future subject-level validation. Figure 10 shows which branches have evidence at each neural- validation tier. Figure 12 renders a local IBC ToM predicted-response contrast on `fsaverage5`; it is a model diagnostic, not a group activation result.

Figure 9. Branch hypotheses map to ROI systems to test next

Dots are predeclared neural target systems for future validation, not observed activation strengths.

	Auditory STG/STS	Language network	Frontoparietal / IPS	TPJ / mPFC ToM	pSTS social motion	hMT+/visual motion	Task timing /control
HCP Language story vs math audio	●	●	●	○	○	○	●
HCP Social social vs mechanical motion	○	○	○	●	●	●	●
IBC Auditory speech vs sound controls	●	○	○	○	○	○	●
IBC Math arithmetic with lexical controls	○	●	●	○	○	○	●
IBC ToM belief vs physical question	○	○	○	●	○	○	●
RSVP Language timing/probe-preserving rerun	○	●	○	○	○	○	●
Biological Motion intact vs scrambled dynamic motion	○	○	○	○	●	●	●

● primary target ○ secondary target ○ not primary

ROI targets are hypothesis targets derived from task constructs and report notes; no ROI-level subject fMRI statistic has been computed here.

Figure 11: ROI target map**Figure 10. Hypothesis outcomes and neural claim status**

The loop found branch-level hypotheses, but most neural/ROI tiers remain future validation work.

	model-feature separation	predicted-response separation	layer-family support	predicted-fMRI fold bridge	observed fMRI ROI/subject
HCP Language robust item-level hypothesis; neural upgrade open	● supported	● supported	● supported	✗ not recovered	□ missing
HCP Social negative in current pipeline; redesign before neural claim	✗ not recovered	✗ not recovered	□ not done	□ not done	□ missing
IBC Auditory candidate; needs acoustic/ROI validation	● candidate	● candidate	□ not done	□ not done	□ missing
IBC Math candidate after lexical fix	● candidate	● candidate	□ not done	□ not done	□ missing
IBC ToM question contrast candidate	● candidate	● candidate	□ not done	□ not done	□ missing
RSVP Language packaging failure; timing-preserving rerun	✗ not recovered	✗ not recovered	□ not done	□ not done	□ missing
Biological Motion motion-aware representation needed	✗ not recovered	✗ not recovered	□ not done	□ not done	□ missing

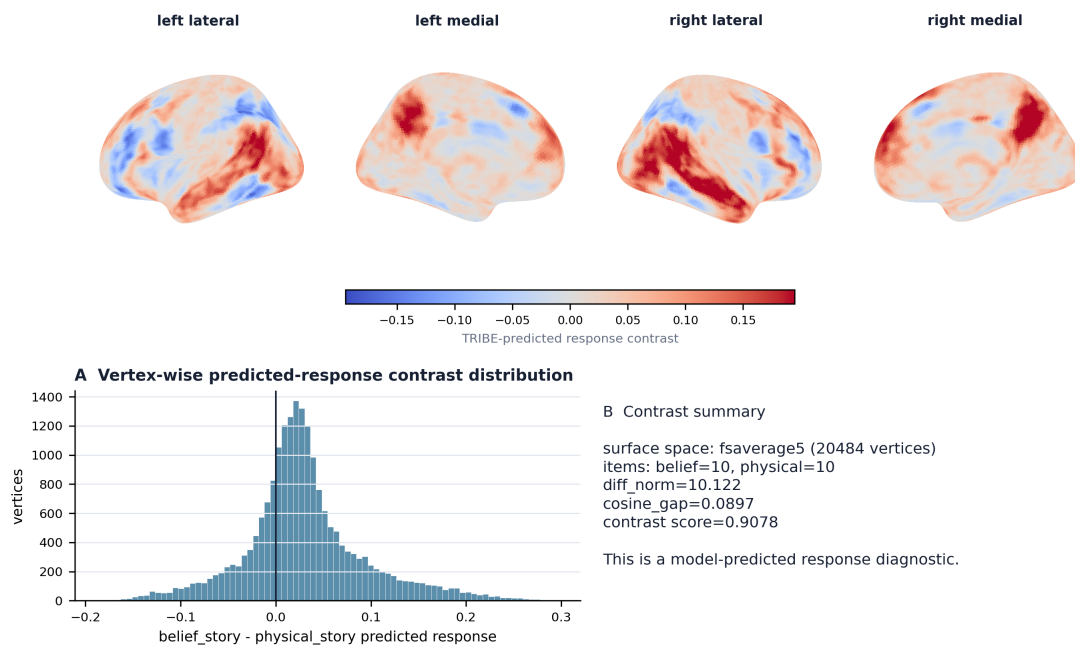
● supported ● candidate ✗ not recovered □ missing/not done

Matrix separates discovered hypotheses from neural confirmation. Missing observed fMRI means no subject-level activation claim yet.

Figure 12: Hypothesis neural-status matrix

Figure 12. IBC ToM predicted fsaverage5 response contrast

TRIBE-predicted belief_story minus physical_story surface response, shown as a diagnostic model-response map.



Predicted response only. This is not observed subject fMRI and is not a statistical activation map.

Figure 13: IBC ToM predicted fsaverage5 response

4. Central hypothesis and registered predictions

The autonomous loop above produces findings; this section converts the strongest finding into a falsifiable scientific hypothesis with predeclared predictions, a falsification protocol, and named alternatives.

4.1 H1 — Registered hypothesis

H1. The story-audio versus math-audio separation observed in late TRIBE encoder attention modules (`encoder.layers.{10,12,14}.1`) reflects a high-level narrative/semantic processing dimension that is also encoded in left-lateralized human language cortex — distinct from low-level acoustic structure, arithmetic-specific computation, and generic task-difficulty.

4.2 Predictions

H1 implies four predictions for observed HCP LANGUAGE fMRI, ordered by stringency.

P1 — Encoding accuracy. In a pre-registered subject-level encoding analysis ($n \geq 20$ HCP S1200 subjects; voxelwise ridge regression with within-subject held-out cross-validation), late-encoder TRIBE features will yield higher held-out prediction accuracy in left-lateralized language ROIs (STG, ATL, IFG, AG) than early-encoder or projector features.

- **Pass criterion:** $\text{mean}(\text{delta_r_bar}) \geq 0.03$ per subject across

language ROIs; paired $t > 2.5$ across subjects, two-sided. Here `delta_r_bar` denotes the per-subject mean of held-out Pearson r improvements of late-encoder features over early/projector features.

P2 — Contrast alignment. Within those ROIs, the story-minus-math contrast in TRIBE-predicted responses will correlate positively with the observed story-minus-math BOLD contrast at the ROI-mean level.

- **Pass criterion:** Pearson $r \geq 0.25$, $p < 0.05$.
- This is the load-bearing test. The current predicted-only fold-stability

bridge gives $r \approx 0.098$, $p \approx 0.442$; if the predicted-vs-observed alignment also fails this threshold, H1 is falsified.

P3 — Acoustic robustness. P1 and P2 hold after residualizing both predicted and observed responses against an extended acoustic covariate set (F0 mean and variance, spectral centroid,

speech rate, MFCC 1-13 summaries), not just the duration / loudness / segment-count set used in the current item-level adjustment.

- **Pass criterion:** the reductions in Δr_{bar} (P1) and contrast

correlation (P2) under extended-acoustic adjustment are $\leq 30\%$ relative to unadjusted.

P4 — Anatomical specificity. P1 holds in language ROIs but not in primary auditory cortex (HCP-MMP A1) or early visual cortex (V1).

- **Pass criterion:**

$\Delta r_{\text{bar}}(\text{language ROIs}) - \Delta r_{\text{bar}}(\text{control ROIs}) > 0$ with paired $t > 2.0$.

4.3 Falsification conditions

- **P1 fails in every language ROI** -> the late-layer separation is not

language-related; H1 reduces to "TRIBE encodes a stimulus-discriminative axis we cannot anchor to brain function."

- **P3 fails (effect collapses under extended acoustic adjustment)** -> the

axis is acoustic, not narrative; the late-layer specificity is incidental.

- **P4 fails (effect equally strong in A1 / V1)** -> the late-layer code is

generic stimulus-energy, not domain-specific.

- **P2 fails (predicted-observed contrast correlation null)** -> the model

captures discriminative features that are not aligned with neural readouts; H1 is unrescued.

Equivocal patterns (e.g., P1 holds but P2 fails) are reported as a sharper redesign instruction, not as confirmation.

4.4 Alternative hypotheses

- **H1a — Acoustic.** The separation is driven by spectral/prosodic

differences between narrators and arithmetic speech. Distinguished from H1 by P3.

- **H1b — Task-difficulty / working-memory.** The separation tracks task

load rather than narrative semantics. Distinguished by adding a difficulty- matched control (e.g., easy story vs hard math; or block-ordering manipulation) — flagged as a future redesign rather than within scope of the immediate decisive experiment.

- **H1c — Compound language axis.** The current evidence cannot distinguish

narrative-specific processing from language comprehension in general. Resolving this requires within-language contrasts (story vs scrambled story; story vs word-list) and is left to a follow-on experiment after the decisive HCP test.

4.5 H2 — Methodological hypothesis

H2. A bounded autonomous loop, when constrained by manifest-delta validation, branch-trajectory hypothesis definition, and signature-specific freezes, produces hypothesis classes (positive / candidate / packaging- failure) whose distribution is informative about both biology and pipeline limits. In particular, the rate of packaging-failure branches is a measurable property of the stimulus-materialization layer, not of the underlying neuroscience.

H2 is consistent with the current campaign (one positive, three candidates, three packaging-failures across two stimulus families). It is not falsified or confirmed by a single campaign. Its empirical content is that — across multiple independent campaigns with different stimulus sets and representations — the packaging-failure rate should track materialization- layer fidelity (e.g., timing preservation, motion-aware features) rather than the underlying scientific domain.

5. The decisive experiment

H1 is decided by a pre-registered subject-level encoding analysis on observed HCP LANGUAGE fMRI.

- **Dataset.** HCP S1200 LANGUAGE task release; subject-aligned BOLD or

grayordinate-space CIFTI; $n \geq 20$ subjects with both story and math runs and meeting the standard HCP motion-exclusion criteria.

- **Design matrix.** Story vs math block regressors at HCP-standard timing,

convolved with a canonical HRF; nuisance regressors per the HCP minimal preprocessing pipeline conventions.

- **Encoding model.** Voxelwise ridge regression; train/test on within-

subject held-out runs; features from late vs early vs projector TRIBE families per §2.5; per-voxel held-out Pearson r as the prediction- accuracy statistic.

- **ROIs.** Language: HCP-MMP STGa/STGp, IFGo/IFGr, ATL (TGd, TGv), AG

(PGa, PGp). Control: A1, V1.

- **Decision rule.** Pass = P1 AND P2 AND P4 hold at predeclared thresholds

(P3 separately as a robustness check). Failure = any of P1, P2, or P4 fails. Equivocal results (e.g., P1 holds but P2 fails) are reported as a sharper redesign instruction, not as confirmation of H1.

- **Preregistration.** This section is the registered prediction. The

analysis script and seed will be locked at the commit that adds the HCP target package to the workspace, and the locked commit SHA will be appended to this report at that time.

6. Discussion

The campaign shows how a brain-encoding model can be used before subject-level validation to prioritize stimulus contrasts. It did not treat model separation as a neural result. Instead, it used TRIBE to decide which contrasts should advance, which needed stronger controls, and which failed because the current stimulus representation was inadequate. This produced three outputs: a robust positive HCP language axis (the empirical basis of H1), pipeline-specific failures that guide redesign (the empirical basis of H2), and operational rules for autonomous scientific discovery.

The strongest scientific result is HCP language. It survived multiple item-level validation tiers, remained significant after available acoustic covariate adjustment, and was strongest in late TRIBE encoder attention representations. This supports H1: the TRIBE model captures a high-level auditory story/math separation in late multimodal representations that is hypothesized to align with left-lateralized language cortex. However, the story-vs-math contrast remains compound. It may reflect narrative semantics, language comprehension in general, acoustic or prosodic differences, arithmetic demand, task structure, or working-memory differences. The current evidence is strong enough to justify the locked subject-level validation experiment of §5, but not strong enough on its own to claim language specificity or observed neural activation; that is what P1–P4 are designed to settle.

The most important negative result is HCP social. The branch did not recover social-animation vs mechanical-motion separation, and the score weakened after a valid follow-up. That is a meaningful pipeline result, but it should not be overinterpreted as a biological null. The more defensible conclusion is that the current stimulus packaging and representation/scoring setup is not sensitive enough to the social-intentional and motion features required by this contrast. The next test should use motion-energy, object-count, trajectory-complexity, and social-intentionality controls, plus video- or motion-aware representations.

The candidate branches show how failures become experimental design rules. Auditory requires larger item blocks and acoustic controls. Math requires lexical, visual-notation, and difficulty controls. ToM requires question-level belief-vs-physical contrasts rather than story-only plateaus. RSVP language requires timing and probe preservation. Biological motion requires dynamic motion-aware features. These are not merely engineering notes; they are the stimulus design principles generated by the discovery loop, and they are the empirical content of H2.

The methodological contribution is a two-level definition of hypotheses. The intermediate unit is the full test record for a candidate contrast: the input stimuli, the score pattern, the follow-up or validation result, and the next-experiment decision. The final unit is a falsifiable scientific hypothesis about the brain, registered only after a branch has survived those checks. This two-level definition prevents two common failure modes: treating no-op follow-ups as scientific progress, and stopping at a high score without ever stating what about the brain we now believe.

7. Limitations

The first limitation is the missing observed fMRI validation. The current workspace does not contain subject/run-aligned HCP LANGUAGE observed target data. Without aligned BOLD, CIFTI, NIFTI, beta, COPE, ZSTAT, or stat-map targets and a stimulus-to-run design matrix, H1 cannot be confirmed or falsified yet. This is exactly what §5 is designed to address.

The second limitation is that the contrast score is a model-space statistic, not a p-value or neural effect size. P-values in this report come only from the explicit permutation validations. Scores across raw predicted responses, residualized features, and layer-feature spaces should not be treated as numerically interchangeable.

The third limitation is confounding in the HCP language contrast. Duration and segment count were well balanced in the held-out sidecar, and loudness was included in residualized checks, but speaker identity, prosody, F0, spectral centroid, speech rate, MFCCs, transcript length, math difficulty, and working-memory demand are not fully controlled. Several of these are explicitly addressed by P3; the remainder will require a follow-up experiment after the decisive HCP test.

The fourth limitation is that many negative branches are packaging- sensitive. RSVP and biological motion likely failed because the current materialization collapsed task-defining temporal or motion structure. HCP social may require motion-aware features and stronger stimulus controls. These branches are best reported as current-pipeline non-recoveries.

The fifth limitation is the strength of pre-registration. The layer-family confirmatory test is anchored to commit 7f1c89b1, which adds both the locked manifest builder and the validator. Strict pre-registration would require an earlier commit isolating the predicted statistic from its evaluation. We treat 7f1c89b1 as a soft pre-registration anchor for the layer-family result, and treat §4 (H1, P1–P4) as the strict pre-registration for the decisive experiment.

The sixth limitation is the use of an AI-generated schematic for Figure 1. Figure 1 was generated with Nano Banana from a structured prompt and does not encode any data; it is conceptual. This is disclosed in the figure legend.

8. Conclusion

The TRIBE stimulus-discovery campaign converts a self-driven discovery loop into one registered scientific hypothesis (H1) and one methodological hypothesis (H2). H1 states that the story-audio vs math-audio separation seen in late TRIBE encoder attention modules reflects a high-level narrative/semantic processing dimension encoded in left-lateralized human language cortex, and is decided by four pre-registered predictions on observed HCP LANGUAGE fMRI (§4–§5). H2 states that the distribution of branch outcomes from a guarded autonomous loop is itself informative about the stimulus-materialization layer, and is decided across future campaigns.

The decisive next experiment is fixed: subject/run-aligned HCP LANGUAGE encoding with the locked late-layer feature family, with predeclared language vs control ROIs and a predeclared pass/fail decision rule. Until that experiment is executed, the registered claim of this report is the hypothesis itself, not a confirmed neural result.

Ethics, data, and code availability

Ethics. All analyses reported here use stimulus-level metadata and stimulus-aligned model representations; no subject-identifying data are reported. Subject-level fMRI targets, when added for the decisive experiment (§5), will be obtained and used under HCP Open Access Data Use Terms with the relevant Data Use Agreement.

Data availability. Compact figure-data tables (CSV/JSON), prediction- row copies, and locked manifest specifications are mirrored under `/data/brain_researcher/research/discovery/docs/operations/`. The HCP and IBC raw stimulus assets are distributed under their respective data-use terms; this report does not redistribute raw stimuli.

Code availability. All analysis, validation, and figure-generation scripts live under `/home/zijiaochen/projects/brain_researcher/scripts/autoresearch/discovery/`. The reproduction commands and ordered execution pack are listed in the Reproducibility section below. The layer-family confirmatory work is anchored at git commit `7f1c89b1`.

Reproducibility and script inventory

This section is intentionally path-heavy so the report remains self-contained. The conceptual and statistical claims above should be read together with this inventory. Paths under `/home/zijiaochen/projects/brain_researcher` are local repository paths. Paths under `/data/brain_researcher/research/discovery` are local evidence/report artifacts. Paths under `/home/ubuntu` are remote TRIBE VM runtime paths that were inspected or used during the validation campaign.

Model and runtime sources

The TRIBE runtime entrypoint inspected on the VM was `/home/ubuntu/tribe_encoding/project/scripts/sweep/run_wave1_pilot.py`. The Brain Researcher wrapper inspected locally was `brain_researcher.services.tools.tribe_tool.TribePredictTool`. The installed TRIBE model implementation inspected on the VM was `/home/ubuntu/miniconda3/envs/tribe/lib/python3.11/site-packages/tribev2/model.py`. The checkpoint was loaded as `facebook/tribev2` with `checkpoint_name=best.ckpt`. The inspected forward path was `aggregate_features -> optional temporal_smoothing -> transformer_forward / encoder -> low_rank_head -> predictor -> pooler`, which is why hooks were placed on projectors, selected encoder modules, final-normalization or head-adjacent features, and predictor-adjacent outputs.

The relevant remote closed-loop root was `/home/ubuntu/tribe_encoding/project/artifacts/closed_loop/persistent_main`. The validation root was `/home/ubuntu/tribe_encoding/project/artifacts/validation/hypothesis_framework_20260426`. The expanded HCP language prediction branch was `hcp_language_expanded20_audio_v5/predictions/wave1_pilot_round_01`. The held-out HCP language prediction branch was `hcp_language_heldout21_audio_v1/predictions/wave1_pilot_round_01`. The layer-family confirmatory branch was `hcp_language_layer_family_confirmatory_v1/layer_features_layer_family_confirmatory_v1`. The predicted-response fold-stability diagnostic was stored as `hcp_language_predicted_fmri_fold_stability_v1/hcp_language_story_vs_math_predicted_fmri_fold_stability.json`.

Execution pack

The execution pack for this report is the ordered set of inputs, scripts, commands, and outputs needed to regenerate the current manuscript and figures. It is a reproducibility pack for the item-level model-feature, predicted-response, layer-family, and reporting artifacts. It is not an observed-fMRI execution pack because the subject/run-aligned HCP LANGUAGE target data are not present.

The pack has three roots. The repository root is `/home/zijiaochen/projects/brain_researcher`. The report/evidence mirror is `/data/brain_researcher/research/discovery/docs/operations`. The figure root is `/data/brain_researcher/research/discovery/docs/operations/figures`. Commands should be run from the repository root after activating the `brain_researcher` environment (Python 3.11; `torch`, `tribev2`, `numpy`, `scipy`, `matplotlib`, `nibabel`, `pandas` per the project lock file). The figure scripts set `MPLCONFIGDIR=/tmp/matplotlib-cache` internally, so they do not require a display server. Regenerating the AI schematic for Figure 1 is optional and requires the local `.env` API configuration; deterministic evidence figures do not require that API path.

The recommended preflight checks are `python -m py_compile scripts/autoresearch/discovery/make_tribe_branch_outcome_landscape.py scripts/autoresearch/discovery/make_tribe_remaining_figures.py scripts/autoresearch/discovery/make_tribe_neural_bridge_figures.py scripts/autoresearch/discovery/make_tribe_representation_figures.py scripts/autoresearch/discovery/make_tribe_predicted_response_surface_figures.py scripts/autoresearch/discovery/render_tribe_latex_report.py` and a `file-exists` check over all image paths embedded in `docs/operations/tribe_stimulus_discovery_paper_report_2026-04-28.md`. The expected image count is 13.

The deterministic figure-generation sequence is: `python scripts/autoresearch/discovery/make_tribe_branch_outcome_landscape.py`; `python scripts/autoresearch/discovery/make_tribe_remaining_figures.py`; `python scripts/autoresearch/discovery/make_tribe_neural_bridge_figures.py`; `python scripts/autoresearch/discovery/make_tribe_representation_figures.py`; and `python scripts/autoresearch/discovery/make_tribe_predicted_response_surface_figures.py`. This sequence regenerates Figures 2–13 from the compact CSV/JSON figure data, local ToM prediction artifacts, and copied HCP-language prediction-row inputs. Figure 1 is regenerated separately with `python scripts/autoresearch/discovery/generate_nanobanana_tribe_schematic.py` followed by `python scripts/autoresearch/discovery/overlay_nanobanana_tribe_schematic_labels.py`, or reused from the existing cleaned PNG if no schematic restyling is intended.

The manuscript-render sequence is: `python scripts/autoresearch/discovery/render_tribe_latex_report.py`, followed by copying the generated Markdown, LaTeX source, and PDF to the `/data` mirror. The primary expected local PDF is `docs/operations/latex/tribe_stimulus_discovery_paper_report_2026-04-28/tribe_stimulus_discovery_paper_report_2026-04-28_latex_template.pdf`. The primary expected mirrored PDF is `/data/brain_researcher/research/discovery/docs/operations/tribe_stimulus_discovery_paper_report_2026-04-28_latex_template.pdf`. The PDF validation checks are `pdftinfo` on the local and mirrored PDFs, and `pdftotext grep` checks for Central hypothesis and registered predictions, The decisive experiment, Reproducibility and script inventory, and References.

The execution pack outputs should be interpreted by evidence tier. Figures 2–7 are branch-outcome and stimulus-redesign figures. Figures 8–10 are neural-validation boundary and target-planning figures. Figure 11 is the TRIBE-layer representation result. Figures 12–13 are predicted-response

surface diagnostics. None of these outputs substitutes for the missing observed subject/run-fold HCP LANGUAGE fMRI target package required by §5.

Manifest and materialization scripts

The manifest and materialization path is recoverable from these repository scripts: `scripts/autoresearch/discovery/manifest_synthesizer.py`; `scripts/autoresearch/discovery/materialize_hcp_ready_runtime.py`; `scripts/autoresearch/discovery/materialize_biomo_runtime.py`; `scripts/autoresearch/discovery/materialize_patch.py`; `scripts/autoresearch/discovery/build_hcp_language_heldout_manifest.py`; `scripts/autoresearch/discovery/build_hcp_language_covariate_sidecar.py`; `scripts/autoresearch/discovery/build_hcp_language_exchangeability_manifest.py`; and `scripts/autoresearch/discovery/build_hcp_language_layer_family_confirmatory_manifest.py`. These scripts define or repair the stimulus manifests, held-out item splits, covariate sidecars, exchangeability strata, and locked layer- family confirmatory manifests used by the reported validations.

Feature extraction and validation scripts

The layer-feature sidecars were generated with `scripts/autoresearch/discovery/extract_tribe_layer_features.py`. The item- level permutation validators were `scripts/autoresearch/discovery/validate_embedding_permutation.py`, `scripts/autoresearch/discovery/validate_embedding_contrast_permutation.py`, and `scripts/autoresearch/discovery/validate_embedding_contrast_covariate_adjusted.py`. The layer validators were `scripts/autoresearch/discovery/validate_layer_feature_contrast_permutation.py` and `scripts/autoresearch/discovery/validate_layer_feature_family_confirmatory.py`. The predicted-response map bridge was checked with `scripts/autoresearch/discovery/validate_predicted_fmri_fold_stability.py`. Together these scripts separate the exploratory branch scores from the permutation-supported, covariate-adjusted, layer-family, and fold- stability evidence tiers reported above.

Controller and loop guardrail scripts

The operational loop changes that make the branch decisions interpretable are captured by `scripts/autoresearch/discovery/proposal_promoter_patch.py`, `scripts/autoresearch/discovery/score_smoothing_patch.py`, `scripts/autoresearch/discovery/state_evolution_patch.py`, `scripts/autoresearch/discovery/oscillation_convergence_patch.py`, `scripts/autoresearch/discovery/zero_score_refute_patch.py`, `scripts/autoresearch/discovery/run_action_executor.sh`, and `scripts/autoresearch/discovery/run_live_watchdog.sh`. These are not independent neuroscience evidence. They document the controller rules that prevented no-op follow-ups, repeated seed clones, missing terminal actions, and score-oscillation artifacts from being mistaken for scientific progress, and are the empirical basis of H2.

Figure and report generation scripts

The figure-producing scripts are `scripts/autoresearch/discovery/generate_nanobanana_tribe_schematic.py`, `scripts/autoresearch/discovery/overlay_nanobanana_tribe_schematic_labels.py`, `scripts/autoresearch/discovery/make_tribe_branch_outcome_landscape.py`, `scripts/autoresearch/discovery/make_tribe_remaining_figures.py`, `scripts/autoresearch/discovery/make_tribe_neural_bridge_figures.py`, `scripts/autoresearch/discovery/make_tribe_representation_figures.py`, `scripts/autoresearch/discovery/make_tribe_predicted_response_surface_figures.py`, `scripts/autoresearch/discovery/make_tribe_discovery_story_plate.py`, and `scripts/autoresearch/discovery/make_tribe_discovery_paper_plate_v2.py`. The report was rendered with `scripts/autoresearch/discovery/render_tribe_latex_report.py`, which uses the Brain Researcher LaTeX assets `src/brain_researcher/assets/latex/report_template.tex.j2` and `src/brain_researcher/assets/latex/scientific_report.sty`.

Figure data and image artifacts

The compact figure data tables are `/data/brain_researcher/research/discovery/docs/operations/figures/data/branch_outcomes.csv`, `/data/brain_researcher/research/discovery/docs/operations/figures/data/hcp_language_evidence.csv`, `/data/brain_researcher/research/discovery/docs/operations/figures/data/layer_scores.csv`, and `/data/brain_researcher/research/discovery/docs/operations/figures/data/manifest_deltas.json`. The remote prediction-row copies used for figure reproducibility are `/data/brain_researcher/research/discovery/docs/operations/figures/remote_prediction_inputs_20260428/hcp_language_expanded20_audio_v5/embedding_rows.jsonl` and `/data/brain_researcher/research/discovery/docs/operations/figures/remote_prediction_inputs_20260428/hcp_language_heldout21_audio_v1/embedding_rows.jsonl`.

The final image roots are `/data/brain_researcher/research/discovery/docs/operations/figures/nanobanana_schematic_textfree_20260427`, `/data/brain_researcher/research/discovery/docs/operations/figures/figure02_branch_outcomes_20260428`, `/data/brain_researcher/research/discovery/docs/operations/figures/remaining_figures_20260428`, `/data/brain_researcher/research/discovery/docs/operations/figures/neural_bridge_figures_20260428`, `/data/brain_researcher/research/discovery/docs/operations/figures/representation_figures_20260428`, and `/data/brain_researcher/research/discovery/docs/operations/figures/predicted_response_surface_figures_20260428`.

Report outputs

The editable report is `/home/zijiaochen/projects/brain_researcher/docs/operations/tribe_stimulus_discovery_paper_report_2026-04-28.md`. The mirrored data-directory copy is `/data/brain_researcher/research/discovery/docs/operations/tribe_stimulus_discovery_paper_report_2026-04-28.md`. The Brain Researcher LaTeX-template source is `/home/zijiaochen/projects/brain_researcher/docs/operations/latex/tribe_stimulus_discover`

y_paper_report_2026-04-28/tribe_stimulus_discovery_paper_report_2026-04-28.tex. The local LaTeX-template PDF is /home/zijiaochen/projects/brain_researcher/docs/operations/latex/tribe_stimulus_discovery_paper_report_2026-04-28/tribe_stimulus_discovery_paper_report_2026-04-28_latex_template.pdf. The mirrored data-directory PDF is /data/brain_researcher/research/discovery/docs/operations/tribe_stimulus_discovery_paper_report_2026-04-28_latex_template.pdf.

Figure Legends

Figure 1. Conceptual schematic

This schematic shows the scientific setup and autonomous discovery loop: stimuli, TRIBE model representations or layers, predicted response evidence, contrast scoring, and branch decisions. It is conceptual only and is not a data figure. The image was generated with the Nano Banana text-free schematic generator from a structured prompt; no data are encoded in the figure. AI-generation is disclosed here per the report's limitations (§7).

Image: /data/brain_researcher/research/discovery/docs/operations/figures/nanobanana_schematic_textfree_20260427/tribe_nanobanana_conceptual_schematic_1_labelclean.png

Figure 2. Branch outcome landscape

Self-driven experiments sort stimulus contrasts into hypothesis classes. The autonomous loop sorted branches into robust positive axes, candidate/noisy signals, and packaging-sensitive failures. The contrast score is an automated separability measure, not a p-value or direct subject-level neural effect size.

Image: /data/brain_researcher/research/discovery/docs/operations/figures/figure02_branch_outcomes_20260428/figure02_branch_outcome_landscape_20260428.png

Figure 3. Branch trajectories

Branch decisions are trajectory-dependent. HCP language freezes, HCP social weakens after follow-up and is killed, and auditory remains noisy rather than cleanly reportable.

Image: /data/brain_researcher/research/discovery/docs/operations/figures/remaining_figures_20260428/figure03_branch_trajectories_20260428/figure03_branch_trajectories_20260428.png

Figure 4. Score decomposition

HCP social did not fail because centroid distance was zero; it failed because directional separation was tiny after a valid follow-up. Only branch states with audited score components are plotted.

Image: /data/brain_researcher/research/discovery/docs/operations/figures/remaining_figures_20260428/figure04_score_decomposition_20260428/figure04_score_decomposition_20260428.png

Figure 5. Condition signature changes

Real scientific follow-ups require manifest-level condition changes. Math, auditory, ToM, and HCP social illustrate corrected or valid deltas.

Image: /data/brain_researcher/research/discovery/docs/operations/figures/remaining_figures_20260428/figure05_condition_signatures_20260428/figure05_condition_signatures_20260428.png

Figure 6. Finding-hypothesis-next-experiment matrix

The discovery loop produces bounded hypothesis classes and concrete next experiments, not only raw branch scores. Hypotheses in this matrix remain bounded by validation status.

Image: /data/brain_researcher/research/discovery/docs/operations/figures/remaining_figures_20260428/figure06_finding_hypothesis_matrix_20260428/figure06_finding_hypothesis_matrix_20260428.png

Figure 7. Stimulus redesign roadmap

Positive, noisy, and packaging-sensitive branches imply different next experiments rather than a single rerun policy. The roadmap is a plan, not evidence that future experiments have already run.

Image: /data/brain_researcher/research/discovery/docs/operations/figures/remaining_figures_20260428/figure07_redesign_roadmap_20260428/figure07_redesign_roadmap_20260428.png

Figure 8. Neural validation ladder

HCP language has strong item-level and model-feature evidence and late-layer support, but the predicted-fMRI fold bridge is not stable and observed subject-level fMRI validation is missing. This is a validation- tier figure, not an observed fMRI activation map.

Image: /data/brain_researcher/research/discovery/docs/operations/figures/neural_bridge_figures_20260428/figure08_neural_validation_ladder_20260428/figure08_neural_validation_ladder_20260428.png

Figure 9. ROI target map

Branch outcomes imply ROI systems to test next, especially auditory and language targets for HCP language and pSTS / TPJ / mPFC / motion targets for social and biological-motion redesigns. Dots indicate future validation targets, not observed activation or ROI-level statistical significance.

Image: /data/brain_researcher/research/discovery/docs/operations/figures/neural_bridge_figures_20260428/figure09_roi_target_map_20260428/figure09_roi_target_map_20260428.png

Figure 10. Hypothesis neural-status matrix

HCP language is supported at model-feature, predicted-response, and layer-family tiers, while subject-level observed fMRI and most branch-level ROI claims remain missing or not done.

Image: /data/brain_researcher/research/discovery/docs/operations/figures/neural_bridge_figures_20260428/figure10_hypothesis_neural_status_20260428/figure10_hypothesis_neural_status_20260428.png

Figure 11. HCP language TRIBE representation profile

The story-audio vs math-audio separation is strongest in late TRIBE encoder attention representations, not raw audio/text projectors or early encoder layers. This is item-level model-feature evidence, not observed fMRI.

Image: /data/brain_researcher/research/discovery/docs/operations/figures/representation_figures_20260428/figure11_representation_profile_20260428/figure11_representation_profile_20260428.png

Figure 12. IBC ToM predicted fsaverage5 response

Local IBC ToM prediction artifacts can be rendered as TRIBE-predicted fsaverage5 response contrasts. This is a model-predicted diagnostic only, not observed fMRI or group activation evidence.

Image: /data/brain_researcher/research/discovery/docs/operations/figures/predicted_response_surface_figures_20260428/figure12_tom_predicted_fsaverage5_response_20260428/figure12_tom_predicted_fsaverage5_response_20260428.png

Figure 13. HCP language predicted fsaverage5 bridge

The two HCP language predicted-response folds can be rendered as fsaverage5 maps, but their story-minus-math contrast maps are not stable across folds. The map-level bridge is negative ($r \sim -0.098$, $p \sim 0.442$), and the figure is not observed subject-level fMRI activation evidence.

Image: /data/brain_researcher/research/discovery/docs/operations/figures/predicted_response_surface_figures_20260428/figure13_hcp_language_predicted_fsaverage5_bridge_20260428/figure13_hcp_language_predicted_fsaverage5_bridge_20260428.png

Provenance

The full chronological provenance ledger remains `/data/brain_researcher/research/discovery/docs/operations/br_autoresearch_discovery_complete_unified_2026-04-25.md`. The main validation plan, runtime audit, and layer-feature feasibility notes remain `/data/brain_researcher/research/discovery/docs/operations/hypothesis_discovery_experiment_plan_2026-04-26.md`. The locked follow-up manifest specifications remain `/data/brain_researcher/research/discovery/docs/operations/locked_followup_manifest_specs_2026-04-26.md`. The final figure directory is `/data/brain_researcher/research/discovery/docs/operations/figures`. The local plotting, validation, materialization, and report-rendering scripts live under `/home/zijiao.chen/projects/brain_researcher/scripts/autoresearch/discovery`.

The HCP subject-level observed fMRI targets are not included in the current artifact set. This report therefore provides enough provenance to reproduce the item-level model-feature, predicted-response, and figure-generation claims, but it does not provide the observed subject/run-fold neural-validation inputs required by §5. That missing target package remains the required input for the next decisive experiment.

References

- d'Ascoli, S., Rapin, J., Benchetrit, Y., Banville, H., and King, J.-R. (2025). TRIBE: TRImodal Brain Encoder for whole-brain fMRI response prediction. arXiv:2507.22229. <https://doi.org/10.48550/arXiv.2507.22229>
- Meta AI. (2026). facebook/tribev2 model card. Hugging Face. The model card documents TRIBE v2 as a multimodal brain-encoding model for video, audio, and text and describes predictions on the fsaverage5 cortical mesh. <https://huggingface.co/facebook/tribev2>
- Barch, D. M., Burgess, G. C., Harms, M. P., Petersen, S. E., Schlaggar, B. L., Corbetta, M., Glasser, M. F., Curtiss, S., Dixit, S., Feldt, C., Nolan, D., Bryant, E., Hartley, T., Footer, O., Bjork, J. M., Poldrack, R., Smith, S., Johansen-Berg, H., Snyder, A. Z., Van Essen, D. C., and the WU-Minn HCP Consortium. (2013). Function in the human connectome: Task-fMRI and individual differences in behavior. *NeuroImage*, 80, 169–189. <https://doi.org/10.1016/j.neuroimage.2013.05.033>
- Van Essen, D. C., Smith, S. M., Barch, D. M., Behrens, T. E. J., Yacoub, E., Ugurbil, K., and the WU-Minn HCP Consortium. (2013). The WU-Minn Human Connectome Project: An overview. *NeuroImage*, 80, 62–79. <https://doi.org/10.1016/j.neuroimage.2013.05.041>
- Glasser, M. F., Coalson, T. S., Robinson, E. C., Hacker, C. D., Harwell, J., Yacoub, E., Ugurbil, K., Andersson, J., Beckmann, C. F., Jenkinson, M., Smith, S. M., and Van Essen, D. C. (2016). A multi-modal parcellation of human cerebral cortex. *Nature*, 536, 171–178. <https://doi.org/10.1038/nature18933>
- Huth, A. G., de Heer, W. A., Griffiths, T. L., Theunissen, F. E., and Gallant, J. L. (2016). Natural speech reveals the semantic maps that tile human cerebral cortex. *Nature*, 532, 453–458. <https://doi.org/10.1038/nature17637>
- Schrimpf, M., Blank, I. A., Tuckute, G., Kauf, C., Hosseini, E. A., Kanwisher, N., Tenenbaum, J. B., and Fedorenko, E. (2021). The neural architecture of language: Integrative modeling converges on predictive processing. *Proceedings of the National Academy of Sciences*, 118(45), e2105646118. <https://doi.org/10.1073/pnas.2105646118>
- Caucheteux, C., and King, J.-R. (2022). Brains and algorithms partially converge in natural language processing. *Communications Biology*, 5, 134. <https://doi.org/10.1038/s42003-022-03036-1>
- Lu, C., Lu, C., Lange, R. T., Foerster, J., Clune, J., and Ha, D. (2024). The AI Scientist: Towards fully automated open-ended scientific discovery. arXiv:2408.06292. <https://doi.org/10.48550/arXiv.2408.06292>
- Romera-Paredes, B., Barekatin, M., Novikov, A., Balog, M., Kumar, M. P., Dupont, E., Ruiz, F. J. R., Ellenberg, J. S., Wang, P., Fawzi, O., Kohli, P., and Fawzi, A. (2024). Mathematical

discoveries from program search with large language models. *Nature*, 625, 468–475. <https://doi.org/10.1038/s41586-023-06924-6>

Fischl, B. (2012). FreeSurfer. *NeuroImage*, 62(2), 774–781. <https://doi.org/10.1016/j.neuroimage.2012.01.021>

Abraham, A., Pedregosa, F., Eickenberg, M., Gervais, P., Mueller, A., Kossaifi, J., Gramfort, A., Thirion, B., and Varoquaux, G. (2014). Machine learning for neuroimaging with scikit-learn. *Frontiers in Neuroinformatics*, 8, 14. <https://doi.org/10.3389/fninf.2014.00014>

Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3), 90–95. <https://doi.org/10.1109/MCSE.2007.55>

Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., van Kerkwijk, M. H., Brett, M., Haldane, A., del Rio, J. F., Wiebe, M., Peterson, P., Gerard-Marchant, P., Sheppard, K., Reddy, T., Weckesser, W., Abbasi, H., Gohlke, C., and Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, 585, 357–362. <https://doi.org/10.1038/s41586-020-2649-2>